



INDIAN INSTITUTE OF TECHNOLOGY
KHARAGPUR

Stamp / Signature of the Invigilator

EXAMINATION (End Semester)

SEMESTER (Spring)

Roll Number

Section

Name

Subject Number

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2

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2

Subject Name

Database Management Systems

Department / Center of the Student

Additional sheets

Important Instructions and Guidelines for Students

1. You must occupy your seat as per the Examination Schedule/Sitting Plan.
2. Do not keep mobile phones or any similar electronic gadgets with you even in the switched off mode.
3. Loose papers, class notes, books or any such materials must not be in your possession, even if they are irrelevant to the subject you are taking examination.
4. Data book, codes, graph papers, relevant standard tables/charts or any other materials are allowed only when instructed by the paper-setter.
5. Use of instrument box, pencil box and non-programmable calculator is allowed during the examination. However, exchange of these items or any other papers (including question papers) is not permitted.
6. Write on both sides of the answer script and do not tear off any page. **Use last page(s) of the answer script for rough work.** Report to the invigilator if the answer script has torn or distorted page(s).
7. It is your responsibility to ensure that you have signed the Attendance Sheet. Keep your Admit Card/Identity Card on the desk for checking by the invigilator.
8. You may leave the examination hall for wash room or for drinking water for a very short period. Record your absence from the Examination Hall in the register provided. Smoking and the consumption of any kind of beverages are strictly prohibited inside the Examination Hall.
9. Do not leave the Examination Hall without submitting your answer script to the invigilator. **In any case, you are not allowed to take away the answer script with you.** After the completion of the examination, do not leave the seat until the invigilators collect all the answer scripts.
10. During the examination, either inside or outside the Examination Hall, gathering information from any kind of sources or exchanging information with others or any such attempt will be treated as 'unfair means'. Do not adopt unfair means and do not indulge in unseemly behavior.

Violation of any of the above instructions may lead to severe punishment.

Signature of the Student

To be filled in by the examiner

Question Number	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											
Marks obtained (in words)				Signature of the Examiner				Signature of the Scrutineer			

Instructions: Answer all SIX questions. Time = 2hrs. Total marks = $6 \times 10 = 60$. Write your answers only in the space provided. Show the solution steps. Answers without explanation will be penalised. The question paper has total 16 pages.

ROUGH WORK

1.(a) The academic module of IITKGP ERP maintains data about the following entities: (a) courses, including number, title, credits, syllabus, and prerequisites; (b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; (c) students, including student-id, name, and program; and (d) instructors, including identification number, name, department, and title. Further, the enrolment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modelled. The assumptions made are: i. a class meets only at one particular place and time, (i) multiple classes may meet at the same place and time. Construct an E-R diagram for the academic module. Indicate the mapping constraints as well as the weak entities in the diagram. [5]

(b) Convert the above E-R diagram in to a relational table schema. Underline the key attribute(s). [5]

2.(a) Given tables $R(\underline{A}, B)$, $P(\underline{A}, B)$, $S(\underline{B}, C)$, $T(\underline{C}, A)$, with the primary keys underlined. State if each of the following statement is true/false. If it is false, provide a counter-example of the relevant tables with one/two tuples, showing that it is false. All the attributes have integer domain. [5]

Statement	True/False	Counter-Example
A is a key for $R \cap P$		
A is a key for $R \cup P$		
A is a key for $R \bowtie S$		
C is a key for $S \bowtie T$		
C is a key for $R \bowtie S \bowtie T$		

(b) Consider the following relations in an international movie database. Write relational algebra statements to retrieve the results of the queries given below. [5]

Actor(actorId, name, nationality, age); Director(directorId, name, nationality)
Film(filmID, title, year, directorID); PerformsIn(actorID, filmID)

(i) ActorID, name, nationality, age, of all actors who have directed a film.

(ii) Titles of all films in which Leonardo DiCaprio and Kate Winslet have co-acted.

(iii) Names of all actors that have performed in a film directed by Christopher Nolan.

(iv) ActorID, name, nationality, age, of all actors who have appeared in the movie titled Inception.

(v) Names of all actors that have played the character of Bruce Wayne, together with the year the corresponding films were released

3. Given the relational tables related to academic course enrolments:

[5x2=10]

Student(Sid, Name, Degree), Course(Cid, Cname, Room, Teacher), Enrolled(Sid, Cid, Marks)

(i) Write SQL query to obtain the names of the courses in which a MTech student or a BTech student are enrolled.

```
SELECT DISTINCT C.Cname
FROM Course C
JOIN Enrolled E ON C.Cid = E.Cid
JOIN Student S ON E.Sid = S.Sid
WHERE S.Degree = 'MTech' OR S.Degree = 'BTech';
```

(ii) Write SQL query to obtain the names of the courses in which a MTech student and a BTech student are enrolled.

```
SELECT Cname
FROM Course
WHERE Cid IN (
    SELECT E.Cid
    FROM Enrolled E JOIN Student S ON E.Sid = S.Sid
    WHERE S.Degree = 'MTech'
)
AND Cid IN (
    SELECT E.Cid
    FROM Enrolled E JOIN Student S ON E.Sid = S.Sid
    WHERE S.Degree = 'BTech'
);
```

(iii) Translate the relational algebra statement to a SQL query:

$\pi_{\text{Room, Teacher}}(\text{Course})$

```
SELECT Room, Teacher
FROM Course;
```

(iv) Translate the relational algebra statement to a SQL query without using the SQL JOIN operator:

$$\pi_{\text{Sid}}(\text{Enrolled}) - \pi_{\text{Sid}}(\sigma_{\text{Marks} < \text{M1}}(\text{Enrolled} \bowtie \pi_{\text{M1}}(\rho_{\text{Marks} \rightarrow \text{M1}}(\text{Enrolled}))))$$

```
SELECT DISTINCT Sid
FROM Enrolled
WHERE Sid NOT IN (
  SELECT E1.Sid
  FROM Enrolled E1,
       (SELECT Marks AS M1 FROM Enrolled) X
  WHERE E1.Marks < X.M1
);
```

(v) Translate the relational algebra statement to a SQL query without using the SQL JOIN operator:

$$\pi_{\text{Name}}(\sigma_{\text{Mark} > 70}(\text{Student} \bowtie \text{Course} \bowtie \text{Enrolled}))$$

```
SELECT DISTINCT S.Name
FROM Student S
JOIN Enrolled E ON S.Sid = E.Sid
JOIN Course C ON C.Cid = E.Cid
WHERE E.Marks > 70;
```

4. Consider the following set F of functional dependencies for the relation schema $R = (ABCDE)$:

$F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$.

[5x2=10]

(a) Compute the canonical cover F_c , for the above set of functional dependencies.

(b) Determine the candidate key(s) of R

(c) Prove that R is not in BCNF

(d) Suitably decompose R into appropriate relation schemas all of which are in BCNF

(e) Is the obtained decomposition dependency preserving? Briefly justify yes or no.

5. Consider the schema $R = (A, B, C, D, E, G, H)$ and the set F of functional dependencies:

$AB \rightarrow CD, D \rightarrow C, DE \rightarrow B, DEH \rightarrow AB, AC \rightarrow DC$

(a) Determine the list of candidate keys for the above schema R .

[2]

(b) Determine the canonical cover for F , along with the explanation of the steps for its generation.

[4]

(c) Use 3NF decomposition algorithm (3NF synthesis algorithm) to generate a 3NF decomposition. [4]

6. Consider the universal relation $R = (A, B, C, D, E, F, G, H, I, J)$ and the set of functional dependencies $F = \{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$.

(a) Determine the candidate key(s) for R ?

[2]

(b) Decompose R into 2NF and then 3NF relations.

[4]

(c) Consider the following table: *student_course_inst(student, course, instructor)*

Student	Course	Instructor
Narayan	Database	Mark
Smith	Database	Naveen
Smith	OS	Ammar
Smith	Theory	Schulman
Wallace	Database	Mark
Wallace	OS	Ahamad
Wong	Database	Omiecinski
Zelaya	Database	Naveen
Narayan	OS	Ammar

Determine the functional dependencies (FDs) from the above table. At present, what is the highest normal form that the table satisfies? Justify your answer with appropriate reasons. [4]